Altruistic Self-Image and Self-Deception: A Non-Monotonic Relationship

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Abstract

Efforts to manage one's self-image frequently result in self-deception. The broader literature indicates a positive relationship between these two concepts. However, this research's theoretical framework suggests that this monotonic relationship does not persist in contexts where motivations conflict. To construct a conflicting setting, a novel experimental design is introduced, incorporating monetary incentives, altruistic preferences, and self-image concerns. This design allows for an approximation of self-image through a proxy, avoiding the bias associated with self-report methodologies. Participants received a stimulus indicating their participation in one of two distinct allocation tasks; one posed a moral dilemma between maximizing charitable contributions and personal gain, while the other did not. Misinterpreting the signal incurred a cost and did not change the assigned task. The findings reveal a non-monotonic relationship between self-image concerns and self-deception, with the direction of the bias depending on the dominant incentive; either personal gain or self-image concerns. This divergence suggests two distinct types of self-deception: profit-driven and image-driven. Profit-driven self-deceivers underestimated altruistic opportunities and exhibited low to moderate levels of altruism and self-image concerns. Conversely, image-driven self-deceivers overestimated scenarios necessitating ethical judgment and belonged to groups with heightened levels of altruism and self-image concerns. Given the contrasting tendencies of these self-deception types, the overall mean bias was null, highlighting the significance of integrating self-image considerations into self-deception research.

1 Introduction

Many people value being a good person, and this feeling is often tied to altruism. Imagine a group of people with six 1-Euro coins in their pockets walking down a street. On any given day, this street is occupied by five homeless individuals spread out along it. As they pass through the street, these people face a choice: donate one Euro, feel altruistic but become poorer; or withhold the donation, feel less altruistic but remain wealthier.

These passersby could follow one of three strategies. Walking down the street, some prioritize money, caring little about feeling altruistic, and reaching the end of the street with full pockets

but no moral satisfaction. Others value their self-image as a good person but also like money. One way to deal with their problem strike a balance in their giving amount that allows them to keep part of their money and feel some moral satisfaction. Or, they could unconsciously adopt a third strategy: self-deception. They notice two homeless people, and give one Euro to each to avoid guilt, while retaining most of their money. This allows them to maintain a positive and high self-image, while keeping their pockets relatively full. A third group, very sensitive to their self-image, seeks to give as much as they can. Not only do they donate one Euro to each homeless person on the street, but they also mistakenly donate one Euro to a man lying on the floor, too inebriated to reach home after a night of heavy drinking. This group of people reaches the end of the street with empty pockets but with an unmatched sense of moral satisfaction.

In all three groups, self-image concerns play a role, either through their absence or their intensity. The same three groups emerge from the experimental results of this paper: people who are unconcerned, profit-driven self-deceivers, and image-driven self-deceivers. The latter group is of particular interest since it reveals that self-image concerns can lead to multi-directional biases, depending not on the structure of preferences, but on their balance. Self-deception is a subcategory of motivated beliefs, which in turn are part of a wider movement in economics: belief-based utility. Molnar and Loewenstein (2022).

The persistence of altruism, even in the absence of social motives, can be explained by a process of self-impression management Murnighan et al. (2001). In a study by Dana et al. (2007), participants decreased their giving when they could deny it to others, thus decreasing the social cost of their decision. They also decreased their rates when they could deny it to themselves, thus lowering the self-image cost of their decision. Similarly, participants in Dana et al. (2006) often decided to incur a 10% reduction in their maximum payout rather than reveal their selfish decision to their partners. In anonymous lying experiments, individuals can choose to be fully honest, fully dishonest, or lie partially, which supports the self-image management perspective of morality-related decision-making Fischbacher and Föllmi-Heusi (2013). Moral balancing is also identified in Ploner and Regner (2013), where participants who cheated to increase their funds, donated from those funds at higher rates. These findings indicate that prosocial behavior is the combined result of a variety of elements. Bénabou and Tirole (2006) model and subsequently locate four motivational sources: altruistic preferences, material self-interest, social image, and self-image concerns. In the current paper, only social-image concerns are excluded.

Having introduced the specific motivated beliefs strategy that will be examined in this paper, it is now time to look at the specific context. In dictator games, the standard economic view once held that dictator-giving should amount to zero. Nevertheless, decades of experimental data have shown that dictator-giving is often positive. Engel (2011) analyzed 616 dictator game treatments and found that the mean dictator giving is 28.35% of the total funds. Yet, as will be discussed in the following paragraphs, much of this altruistic behavior comes with strings attached. Social preferences are complex, and varied, and can be shaped by a variety of factors (for a review, see Fehr and Charness 2023). In the remainder of this section, I will focus on the different side-effects that motivated reasoning has on prosocial behavior.

In research exploring the relationship between self-deception and altruism, I must mention the influential study by Dana et al. (2007), which was successfully replicated in Larson and Capra (2009). In "Exploiting Moral Wiggle Room", the authors found that altruistic choices decrease when there is plausible deniability or uncertainty, with participants avoiding costless information that could resolve the uncertainty. Note that costless information avoidance contradicts the traditional economic view of information as a decision-enhancing input. Grossman and Van der Weele (2017) provide further experimental evidence of willful ignorance in social decisions, framing it in the context of self-signaling.

In our efforts to decrease our altruism, we may also distort our beliefs about others. Ambiguity surrounding the preexisting fund allocation of another individual can be interpreted in a self-serving manner, leading to reduced altruistic choices Haisley and Weber (2010), and Di Tella et al. (2015) show that our beliefs regarding the corrupt behavior of others increase when that allows us to act egotistically. However, both the experiment conducted by Ging-Jehli et al. (2020) and its replication, done by Di Tella et al. (2015), found no experimental evidence of absolute strategic cynicism. On the contrary, participants in the experiments appeared to hold excessively positive views regarding the behavior of others in absolute terms. Where Ging-Jehli et al. (2020) agrees with the previous research is the existence of relative strategic cynicism. Specifically, individuals who had a lot to gain by acting egoistically tended to convince themselves of others' bad behavior, and individuals who had little power to act egoistically tended to convince themselves that the other would behave altruistically.

This study contributes to the understanding of self-deception in decision-making, particularly in contexts involving altruism and self-image concerns. The participants in my experiment made allocation decisions between themselves and a charity in two different scenarios. In one scenario, they could maximize both their own payout and that of the charity at no cost (*win-win*). In the other scenario, maximizing the charity's payout necessitated a reduction in their own payout (*moral conflict*). Participants received a noisy signal to inform them of the scenario they were in. Identifying the signal is irrelevant to determining which is the active scenario, participants had a monetary incentive to correctly identify it.

The experiment is designed to meet all three conditions required to separate motivated cognition from heuristics (Bénabou and Tirole, 2016), and all four conditions necessary for this motivated cognition to be defined as self-deception (Gur and Sackeim, 1979). Nevertheless, I attempt to refine the first condition of Bénabou and Tirole (2016) regarding the directionality of motivated cognition. The bias that the participants in this research show maintains a directionality, but the direction of the bias differs based on the magnitude of their beliefs rather than their type. In contrast to Ging-Jehli et al. (2020), this difference in direction is not a result of a change in the belief at stake. Ging-Jehli et al. show that, in a relatively similar setting, individuals may reverse the direction of their beliefs about others depending on which belief about themselves is at stake. When the belief involved is fairness, and their actions do not align with this belief, individuals tend to convince themselves that their counterpart acted unfairly as well, therefore deserving the mistreatment. When forced to act virtuously, the belief at stake

becomes whether they are being exploited. Consequently, they may convince themselves that the other acted virtuously and did not take advantage of the situation. In the present research, the belief at stake—altruism—remains the same; however, the direction of the bias changes depending on the magnitude of that belief. Moreover, the groups that follow these directions are identifiable. This represents a significant contribution of my experimental research, because it implies that people with similar beliefs and utility structures can display opposing biases in the same context. These opposing biases result in sub-optimal decisions and loss of welfare.

This study's experimental design enables the investigation of opposing beliefs by measuring proxies of beliefs in a self-signaling setting with conflicting motivations. The categorization of participants into distinct belief groups is a significant novelty of this research. In addition, the experimental design also allows for thorough testing of previous findings. It incorporates various levels of ambiguity, as well as assessments of reaction time.

The study of reaction time poses several challenges Spiliopoulos and Ortmann (2018). However, in this experiment, reaction time can serve as an indicator of self-deception. If less information is required to accept belief-confirming signals Ditto and Lopez (1992), we can expect participants to have shorter reaction times when identifying their preferred outcome. Since the favorable outcome varies across belief groups, reaction times should reflect these differences.

Thanks to the novel experimental design of this research, I can investigate self-deception from multiple perspectives. The most important contribution comes from the measurement of a belief proxy and the categorization of participants into distinct belief groups. This partitioning reveals opposing biases that arise from the same type of belief—a phenomenon that has not been previously documented in the literature.

2 Experimental Design[1]

This experiment explores the relationship between self-deception and altruistic behavior in an ambiguous context. Participants are presented with two scenarios: *win-win* and *moral conflict*. In the win-win scenario, participants can maximize both personal and charity payouts, while in the "moral conflict" scenario, maximizing personal gain reduces charity payout. The charity chosen is "Give Directly", a charity that provides cash directly to impoverished individuals. The choice of Give Directly is a result of an effort to avoid any prior biases regarding the cause of a charity.

The decision-making process involves choosing between actions A and B (Table 1) in one of the two scenarios. Barring any unforeseen combinations of preferences, the win-win scenario is the preferable one, since it maximizes everyone's payout. In the "moral conflict" scenario, more complex combinations of preferences play a role. For someone to give up 2 Euros by choosing action B over action A in the conflicting scenario, they must have some altruistic preferences. Of course, an individual may possess altruistic preferences that are not strong enough to justify losing 2 Euros. Finally, action B in the conflict scenario is designed to be the most efficient

[1] **Click here to see this experiment's instructions.**

Table 1: Scenarios

Win-Win			Moral Conflict	
Give Directly	You		You	Give Directly
5	7	A	7	1.5
1.5	5	B	5	5

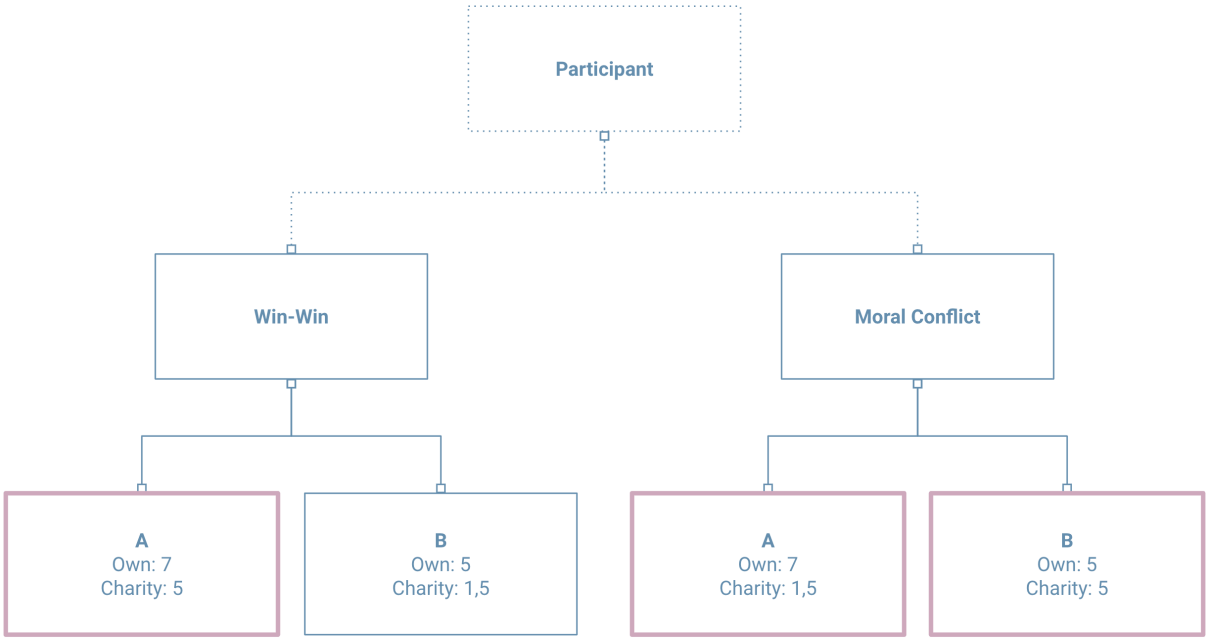

Figure 1: Experiment Decision Tree

choice, so that the more selfish option (action A) is less appealing.

Ambiguity is of vital importance in the manifestation of self-deception, and the experimental setting introduces it with a stimulus signal that indicates the active scenario. Before each decision, participants are shown a screen displaying 84 dots for 250ms. The dots are unequally spread between the right and left sides of a diagonal line unequally. At the beginning of each trial block, participants are informed about the interpretation of the stimulus. During one block, seeing more dots on the left side of the diagonal line means that the active scenario is the win-win scenario, while during the other, it means that the active scenario is the morally conflicting one. The ambiguity level is determined by the difference in the number of dots on each side (Figure 2). For example, if more dots on the right represent the moral conflict scenario, stimuli marked in red refer to the moral conflict scenario and the one in green to the win-win scenario.

To ensure truthful stimulus identification reports, participants are incentivized to accurately report their perceived scenario with 1 Euro. Participants must identify the active scenario based on the stimulus, knowing that their response does not influence the actual scenario they are in. For example, if there are more dots on the right side of the diagonal line, i.e. signifying the

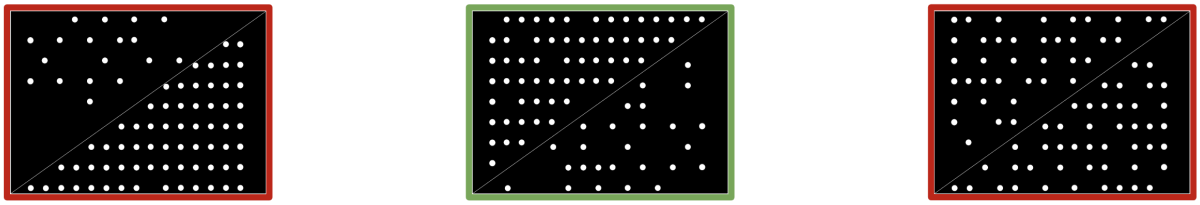

Figure 2: Dot Stimuli Examples

moral conflict scenario, choosing action A always gives 7 Euros and 1.5 Euros to the charity, regardless of which side they believe has more dots. Let's assume that the screen presented to a participant is the one in Figure 2 (a), with more dots on the left of the line; this would mean that the participants are making decisions in the win-win scenario. If they correctly report "left" as the side with more dots, they receive 1 Euro as a bonus. If they report "right" as the side with more dots, they do not receive the 1 Euro bonus. Note, that even if they incorrectly choose "right" as the answer, the active scenario remains the same (win-win). The signal in Figure 2 (a) is stronger (44 more dots on the left) than in Figure 2 (b) (16 more dots on the right), therefore, there is more room for self-deception in the second case.

The subjects of the experiment complete two blocks of trials, with the only difference between them being the reversed interpretation of the stimuli. Each session includes 48 trials. The sequence of events for each trial is stimulus presentation, pattern identification, and allocation decision. At the end of the experiment, participants are incentivized to report the percentage of correct pattern identifications. This last assessment is used to gauge potential awareness of self-deception. Finally, some sociodemographic questions are collected.

In summary, most participants would prefer to see a stimulus that signifies a scenario where both parties maximize their payouts. However, their stimulus identification does not affect the active scenario, and a correct identification is monetarily incentivized. At the same time, as the stimulus signal grows stronger, unconsciously manipulating information becomes harder.

3 Theoretical Framework

3.1 Modeling Approach

The model presented below attempts to describe a morally conflicting decision; where the source of the moral conflict is an altruistic self-image. Similarly to Engelmann et al. (2024) and its model of anticipatory anxiety, I combine a few different approaches to the modeling of self-deception to describe the current setting. As explained in previous sections, the task that the participants undertake fulfills all four conditions of self-deception that were laid out in Gur and Sackeim (1979). The self-deception this experiment deals with is a result of an attempt at self-signaling Mijović-Prelec and Prelec (2010). Namely, individuals interpret a signal and, based on their interpretation, the self-image value of their later decision changes. So, in addition to the material payout, a self-image value is added to the expected utility function. Since

individuals might also have genuine altruistic preferences, these preferences are also included in the expected utility function. Therefore, three out of the four elements of prosocial behavior outlined in Bénabou and Tirole (2006) are included (altruistic preferences, material self-interest, and self-image concerns), while only social-image concerns are left out. Ambiguity is important when it comes to self-deception; to account for it, I follow the approach of Bénabou and Tirole (2002) by including a cognitive cost of self-deception, which increases as the context becomes less ambiguous. Finally, in line with the optimal expectations model of Brunnermeier and Parker (2005), this model treats beliefs as objects of choice, though, in an indirect manner. Specifically, individuals do not choose their beliefs but the magnitude and direction of their bias. This final modification of optimal expectations does not significantly change the predictions of the model; however, it allows for a clearer and more interpretable equilibrium which highlights the directionality of self-deception.

3.2 The Model

R and L refer to the dot identification side (right/left). Therefore, the utility derived by correctly identifying "right" as the side with the most dots for someone who always chooses action A no matter which scenario they are in is:

$$U_R[(R, A), (L, A)|R]$$

Self-deception can present itself in individuals with heterogeneous characteristics and preferences, therefore, we must study the sets of strategies that encompass these differences. It is also of interest to study the cases where individuals choose B in one of the two scenarios, $U_R[(R, B), (L, A)|R]$. The cases where the left side has more dots are designed to be symmetrical, so only one side is included in the analysis. In the remainder of this section, I will first describe the model using the $[U_R[(R, A), (L, A)|R]$ strategy profile as an example and derive the self-deception equilibrium for this strategy. Then, I derive the self-deception equilibrium for those individuals who choose B in one of the two scenarios ($U_R[(R, B), (L, A)|R]$). The main assumption of this analysis is that individuals prefer action A (7 Euros for them and 5 for the charity) in the win-win scenario over action B (5 Euros for them and 1.5 for the charity) in the same scenario. Finally, given the previous assumption, I provide the condition that separates the strategies based on individual characteristics.

3.2.1 Always choosing A: $U_R[(R, A), (L, A)|R]$

In the present model, the object of choice is not the belief, but the bias (ω) of the individual. Bias skews the probability that the individual assigns to the scenario to maximize the expected payout. p denotes the initial perceived probability assigned to the scenario; m denotes the correct stimulus identification reward. ΔY denotes the difference in charity payout between choosing A in the two different scenarios; ΔX denotes the difference in personal payout between

choosing A in the two different scenarios; α denotes the individual's altruism and determines how much ΔY affects the expected utility. By design, $\Delta X = 0$ when moving from (R, A) to (L, A).

$$(p + \omega) \cdot (m + \alpha \cdot \Delta Y + \Delta X)$$

This first component represents the material self-interest and altruistic preferences of the individual. It includes the potential monetary reward (m) of the participant in case they identify the scenario correctly, the monetary reward of the participant according to the choice they made (ΔX) in the scenario they are in and the monetary reward of the charity (ΔY), which also depends on the choice made by the participant in the scenario and is weighted by a true altruism factor (α). The expected utility of the first component depends on the probability that the participant assigns on the (ω). So far, altruistic preferences and material self-interest have been included, the only prosocial behavior component missing is self-image concern. Nevertheless, before self-image concerns can be added, some limits to self-deception must be included.

As is evident from the above expression, an individual can always increase (ω) to increase their expected utility. Of course, that is unrealistic, since it implies that, if desired, no amount of unambiguous information would ever reach consciousness. To solve this problem, and to insert a role for ambiguity in the model, a cognitive cost of biased thinking $(\lambda(s))$ is added. The cognitive cost increases with the strength of the signal (s). When the signal (s) is low, ambiguity is high and the cognitive cost of self-deception is low. When ambiguity is low, the cost limits bias and incentivizes the subject to bring ω to 0.

$$-\lambda(s) \cdot \omega^2$$

In simple terms, the cognitive cost shows that when evidence is too strong, it is very difficult to ignore or misinterpret the evidence. In the context of the experiment, this term indicates that self-deceiving when there are 40 more dots on one side is harder than when the difference is only 8.

Subsequently, I am adding the third element of prosocial behavior, self-image concerns (σ), which capture the increase in utility derived by a positive self-image update related to the individual's choice between A and B, or the decrease of it by a negative update. c is equal to 1 when it signifies the moral conflict scenario and 0 when it signifies the win-win scenario.

$$-\sigma \cdot [c \cdot (p \cdot \omega) + (1 - c)(-(p \cdot \omega)$$

Some individuals are more sensitive to their self-image than others, this component includes a factor that represents that. When the scenario represented by a stimulus with more dots on

the right is morally conflicting, positively biasing their belief in the scenario implies a larger reduction of utility, therefore, some individuals might add a negative bias to their belief to avoid the cost. Likewise, if the scenario is non-conflicting, the individual might increase their chosen belief through the bias term (ω). The size of the cost of changing these beliefs depends on each participant's self-image concerns, or in other words, on their self-image sensitivity to their actions. In the absence of self-image concerns, the cognitive cost is assumed to be strong enough to bring (ω) to zero.

Put together, the expected utility is:

$$U_R[(R,A),(L,A)|R] = (p+\omega)\cdot(m+\alpha\cdot\Delta Y+\Delta X)-\lambda(s)\cdot\omega^2-\sigma\cdot[c\cdot(p+\omega)+(1-c)(-(p+\omega)] \quad (1)$$

Maximizing the expected utility with respect to parameter ω, yields:

$$\omega^* = \frac{\alpha\cdot\Delta Y - 2\cdot c\cdot\sigma + m + \sigma}{2\cdot\lambda(s)}$$

The amount of self-deception can be measured by the difference in the belief that the answer is correct between the cases that the right side scenario is the win-win scenario ($c = 0$) and those that it is the moral conflict scenario ($c = 1$). So, self-deception in this case is defined as $\omega(0) - \omega(1)$.

In this model, the existence of the moral conflict scenario is implicit and related to the material payout. As a result, when $c = 1$, Y_R is negative ($\underline{\Delta Y}$) (which is what makes R the moral conflict scenario), and when $c = 0$, Y_R is positive ($\bar{\Delta Y}$).

In this experiment, $\bar{\Delta Y}$ = - $\underline{\Delta Y}$, therefore, the equilibrium amount of self-deception is:

$$SD_{AA} = \omega^*(0) - \omega^*(1) = \frac{\alpha\cdot\Delta Y + \sigma}{\lambda(s)} \quad (2)$$

3.2.2 Choosing B in one scenario: $U_R[(R,B),(L,A)|R]$

The expected utility now takes the following form:

$$U_R[(R,B),(L,A)|R] = (p+\omega)\cdot(m+\Delta X) - \lambda(s) + \omega^2 + \sigma\cdot[c\cdot(p\cdot\omega) + (1-c)\cdot(-(p+\omega))]$$

The difference with the previous strategic profile is that now $\Delta X > 0$ and $\Delta Y = 0$ by design, and self-image concerns are not costly, but profitable (sign reversal), since choosing B when facing a moral conflict scenario is the altruist choice, providing a positive self-image.

Following the same steps as above we find:

$$\omega^* = \frac{\Delta X + 2\cdot c\cdot\sigma + m - \sigma}{2\cdot\lambda(s)}$$

And:

$$SD_{BA} = \omega^*(0) - \omega^*(1) = \frac{\Delta X - \sigma}{\lambda(s)} \quad (3)$$

3.2.3 Choice under a known moral conflict scenario

It is also interesting to study the conditions under which someone chooses B over A. In the win-win scenario, the utility derived by choosing A is always larger than choosing B since both the individual and the charity maximize their payout, therefore, in this section, the focus is the moral conflict scenario.

Under the assumption that the moral conflict scenario is known, $p = 1$; therefore, individuals choose B in the moral conflict scenario when:

$$U(A|Conflict) < U(B|Conflict)$$
$$=> X_A + Y_A \cdot \alpha - \sigma < X_B + Y_B \cdot \alpha + \sigma$$

$$<=> \frac{\Delta X - 2 \cdot \sigma}{\Delta Y} < \alpha \tag{4}$$

Rearranging the terms to reflect material interests on the left and altruistic preferences along with self-image concerns on the right provides a clearer picture:

$$<=> \Delta X < \alpha \cdot \Delta Y + 2 \cdot \sigma$$

A larger material payout decreases altruistic choices, while altruism and self-image sensitivity both increase altruistic choices. Given that the values of ΔX and ΔY are known, we can simplify further:

$$=> 1.75 \cdot \alpha + \sigma > 1$$

Note that - although the altruism factor is additionally weighted - a sufficiently high level of self-image concerns could also make an individual choose the altruistic outcome.

4 Model Analysis

4.1 Factors of self-deception

4.1.1 Self-Image of Altruism Sensitivity

Altruism self-image sensitivity (σ) increases self-deception when both altruism (α) and σ are low enough that inequality 4 is not satisfied. If both of these terms are sufficiently high—and the inequality is satisfied—σ decreases self-deception and can even make it negative when it becomes very large.

Individuals with characteristics that do not satisfy inequality 4 prefer to hurt the charity than to lose money in the morally conflicting scenario. If σ is low, these individuals' self-image does not get hurt by their selfish actions, therefore, they do not attempt to self-deceive. Nevertheless, if σ is high enough to be costly for these individuals, but not high enough to make them choose the altruistic outcome, they will attempt to self-deceive to protect their self-image.

On the other side, when inequality 4 is satisfied, individuals choose the altruistic outcome. However, not all of them with the same excitement. Those individuals who only barely satisfy the inequality have a low σ and would prefer to just be in the win-win scenario and make more money. As σ increases, the utility from acting altruistically also increases, making the morally conflicting scenario more appealing. In a more extreme scenario, a high σ could provide satisfaction from acting altruistically that is large enough to make the morally conflicting scenario preferable for these individuals. These individuals might attempt to self-deceive in the opposite direction, resulting in negative self-deception.

4.1.2 Altruism

Similarly to self-image concerns (σ), altruism (α) exhibits a complex relationship with self-deception. When inequality 4 is not satisfied, a higher level of altruism does increase self-deception. These individuals choose the non-altruistic outcome and will not self-deceive when their level of altruism is low. However, when their altruism is sufficiently high, the win-win scenario allows them to significantly increase their expected utility, so they might attempt to self-deceive towards that scenario. When both α and σ are large enough, inequality 4 is satisfied and individuals prefer the altruistic choice in the morally conflicting scenario. Altruism does not affect their self-deception levels anymore because of the payout structure of this experiment ($\Delta Y = 0$).

4.1.3 Monetary incentive and payouts

While existing theoretical models suggest that a higher monetary incentive reduces self-deception levels, the present model predicts that the size of the monetary incentive is not relevant. This result is in line with previous experimental findings (Engelmann and Pessoa, 2014; Engelmann et al., 2009).

By design, ΔY is only relevant for SD_{AA} (2) and ΔX is only relevant for SD_{BA} (3). In both cases, an increase in either ΔY or ΔX increases the level of self-deception towards the win-win scenario. More specifically, individuals who tend to choose the altruistic option in the morally conflicting scenario, are tempted to self-deceive towards the win-win scenario when ΔX increases. Individuals that tend to choose the non-altruistic option in the morally conflicting scenario already maximize their monetary payout, therefore, are only tempted to self-deceive towards the win-win scenario if ΔY increases, and as long as they have a positive α.

4.1.4 Cost of Self-Deception

The cost of self-deception ($\lambda(s)$) limits self-deception, as expected. The more ambiguous the setting, the lower the cost and the higher the absolute level of self-deception.

4.2 Measuring Altruism and Self-image

The main goal of this research is to disentangle the relationship between altruism and self-deception. Figure 3 shows the expected relationship that altruism and self-image sensitivity have with self-deception. Both characteristics increase (profit-driven) self-deception until inequality 4 is reached (strategy profile: Always A); after that point (strategy profile: A and B), true altruism ceases to play a role, and self-image sensitivity decreases profit driven self-deception. A strong enough self-image sensitivity could create self-deception in the opposite direction (image-driven).

Directly measuring self-image (α) is not feasible. To address this, a function $\Phi(\alpha, \sigma)$ is assumed, describing an individual's exhibited altruism in unambiguous scenarios without external pressure. Φ increases with both true altruism and self-image sensitivity. It quantifies altruistic choices where self-deception is not possible. Ambiguity is experimentally increased (increase in s), making the internal self-deception cost ($\lambda(s)$) prohibitively high. To achieve this, 8 of 48 trials included strong signals with little room for identification errors. Half of these trials signify a morally conflicting scenario. The share of altruistic choices in those 4 trials represents the Φ for each individual.

Although inequality 4 implies strict and constant values of α and σ for each individual, in practice individuals might slightly vary these values implicitly in a repeated trial setting, therefore, both of these factors should be seen as averages and individual trials might deviate from these averages. For instance, an individual for whom inequality 4 holds on average might choose the altruistic outcome 3 times out of the 4 unambiguous trials but deviate once by choosing the non-altruistic outcome. An individual with particularly low values of α and σ will have $\Phi = 0$ and individuals with very high values will have $\Phi = 1$. For individuals with intermediate values of α and σ, $\Phi \in [\frac{1}{4}, \frac{2}{4}, \frac{3}{4}]$. For simplicity, these groups will be referred to as follows:

- $\Phi = \frac{0}{4} \rightarrow$ Selfish
- $\Phi = \frac{1}{4} \rightarrow$ Partially selfish
- $\Phi = \frac{2}{4} \rightarrow$ Neutral
- $\Phi = \frac{3}{4} \rightarrow$ Partially altruist
- $\Phi = \frac{4}{4} \rightarrow$ Altruist

Although (α) and (σ) are not necessarily correlated, they are both positively correlated with Φ. The reversal of the relationship between (σ) and self-deception after inequality 4 provides a unique opportunity to observe whether self-image concerns dominate. Figure 4 shows the relationship between the measurable Φ and self-deception.

4.3 Econometric Model

A random-effects probit regression model will be used, with the probability of mistake as the dependent variable. The independent variables are the strength of the signal ($\lambda(s)$), a binary

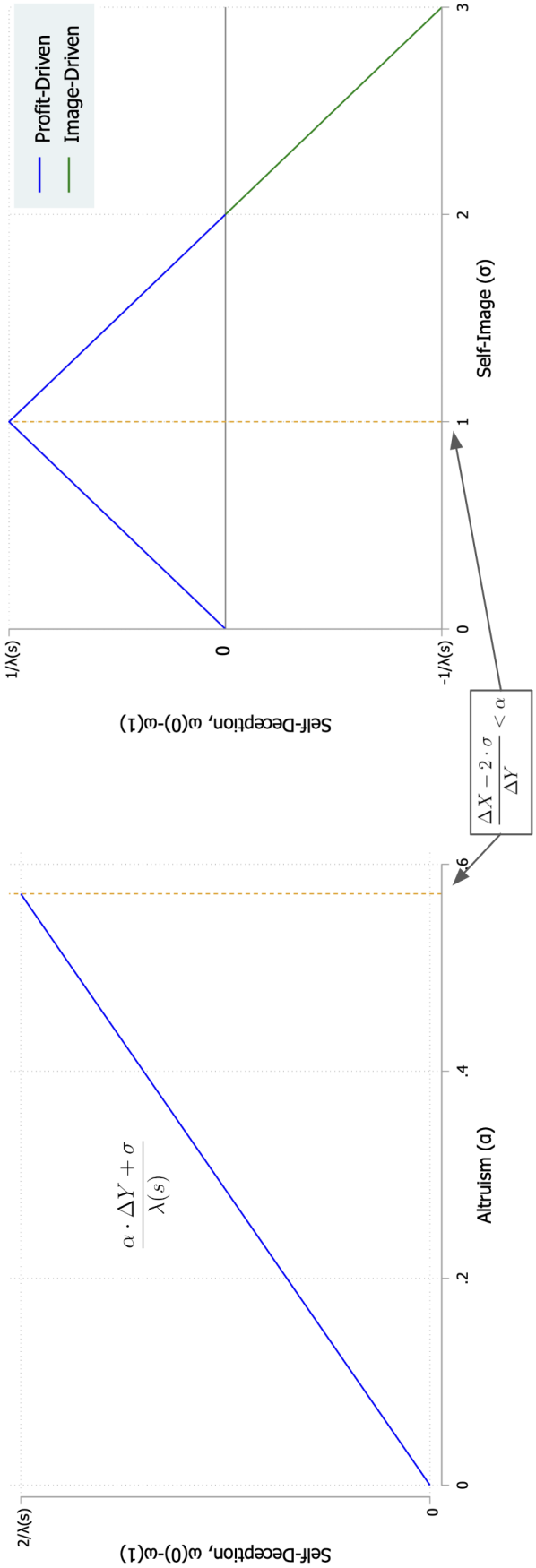

Figure 3: Self-deception and altruism α self-image σ

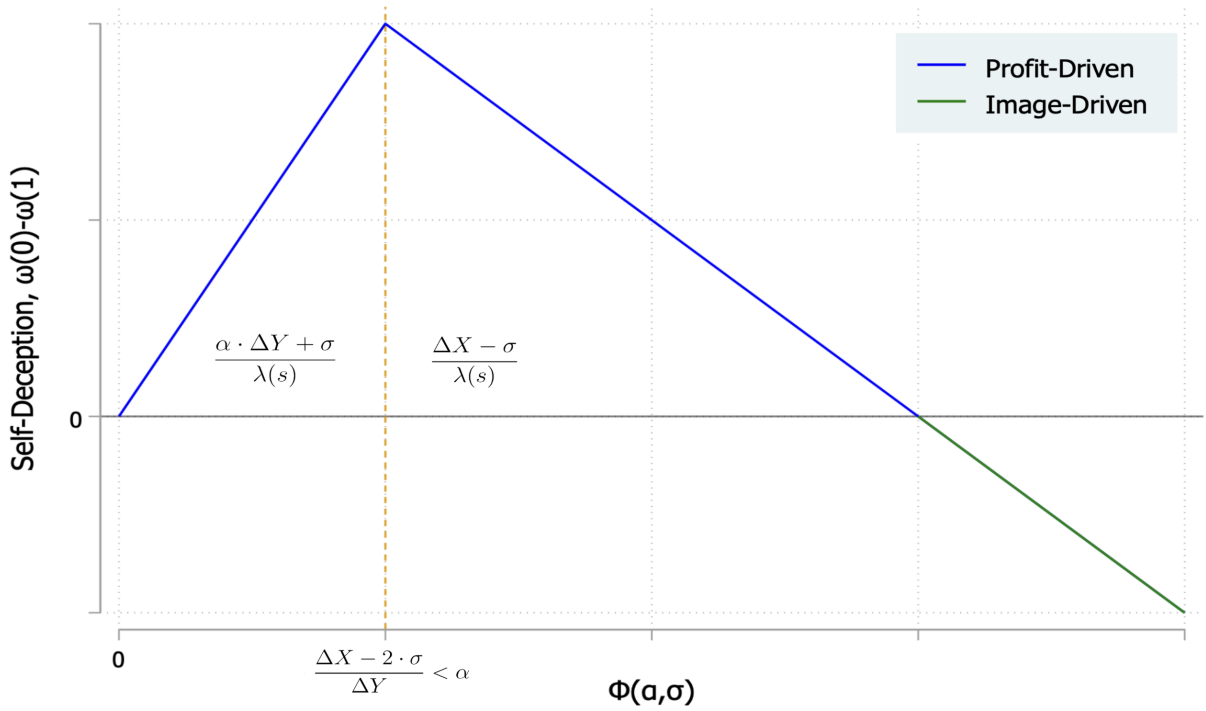

Figure 4: Self-deception and proxy of self-image and altruism Φ

variable indicating whether the trial refers to the moral conflict scenario (C), and the proxy of altruistic preferences, Φ. An interaction term between $\lambda(s)$ and C is added to capture the effect that ambiguity has specifically on biased mistakes. Finally, interaction terms between C and Φ are included. The econometric model takes the following form:

$$
\begin{aligned}
Y_i = \beta_0 + \beta_1 \cdot \lambda(s) + \beta_2 \cdot \ \mathrm{C} + \beta_3 \cdot \ \mathrm{C} \cdot \lambda(s) \\
+\beta_4 \cdot \Phi_{\frac{1}{4}} + ... + \beta_7 \cdot \Phi_{\frac{4}{4}} \\
+\beta_8 \cdot \mathrm{C} \cdot \Phi_{\frac{1}{4}} + ... + \beta_{11} \cdot \mathrm{C} \cdot \Phi_{\frac{4}{4}} + \epsilon_i
\end{aligned}
$$

4.4 Hypotheses

Hypothesis 1, There is self-deception among participants.

A self-deceiving mistake is a mistake that happens when $C = 1$, namely, when the scenario is morally conflicting.

H_0: $b_2 = 0$
H_1: $b_2 > 0$

Hypothesis 2, Ambiguity increases self-deception.

Mistakes are expected to generally increase with the ambiguity of the stimuli since the task becomes objectively harder. Our model also predicts that self-deception (W) increases with the

ambiguity of the stimulus, s. In the econometric model, this is represented with the interaction term between C and $(\lambda(s))$.

H_0: $b_3 = 0$
H_1: $b_3 > 0$

Hypothesis 3, Self-deception peaks in neutrals ($\Phi(\alpha, \sigma) = \frac{2}{4}$).

Similarly to the second hypothesis, we are interested in the set of interaction terms between the moral conflict scenario, C, and the proxy of altruism, Φ. Φ will be positively correlated with self-deception up to $\Phi = \frac{2}{4}$ and negatively from $\Phi = \frac{3}{4}$ to $\Phi = \frac{4}{4}$. Neutrals are expected to exhibit the highest levels of self-deception, while perfectly selfish and perfectly altruist participants will exhibit the lowest amounts of self-deception.

H_0: $b_8 = ... = b_{11} = 0$
H_{1_A}: $b_8 > 0$
H_{1_B}: $b_9 > 0$
H_{1_C}: $b_{10} > 0$

5 Results

Two days of sessions were conducted at Pompeu Fabra University's Behavioral Experimental Sciences Laboratory (BESLab) and three days of sessions were conducted at the Behavioral Experimental Science Laboratory of the University of Barcelona. One of the session days had to be excluded from the analysis due to excessive rational choice violations (see Appendix A2). This section presents the experimental results from 190 participants (106 female, 80 male, 4 who did not identify with the former two). The average age of the participants was 21.16 years and the majority of them (81.59%) were undergoing their undergraduate studies. Finally, 6 Φ groups were identified and included 44 Selfish, 12 Partially Selfish, 15 Neutrals, 20 Partially Altruist, 90 Altruist, and 9 Virtue Consumers. The last group refers to participants who even made a mistake in the easiest 8 trials, resulting in choosing the altruistic outcome one additional time compared to the expected maximum. Instead of adding this group with the altruists, the group has been isolated due to its outlier behavior.

5.1 Descriptive Results

The stimulus task produced a total error rate of 6.18%, with the majority of errors, 84.9%, committed in the hardest 3 levels of the 12 levels of difficulty of the task. The easiest two levels of the task are excluded from the mistake analysis to avoid endogeneity since those two levels are used to build the proxy of altruism and self-image, Φ. Figure 5 shows the average mistake rates per difficulty level. The stronger a signal is, the easier the trial. As can be seen, the error rates remain close to zero for signals larger than 20. Table 2 compares the number of mistakes

between the hardest 20 and the easiest 20 trials. There were almost 9 times more mistakes in the harder trials (12.66%) than in the easier trials (1.42%).

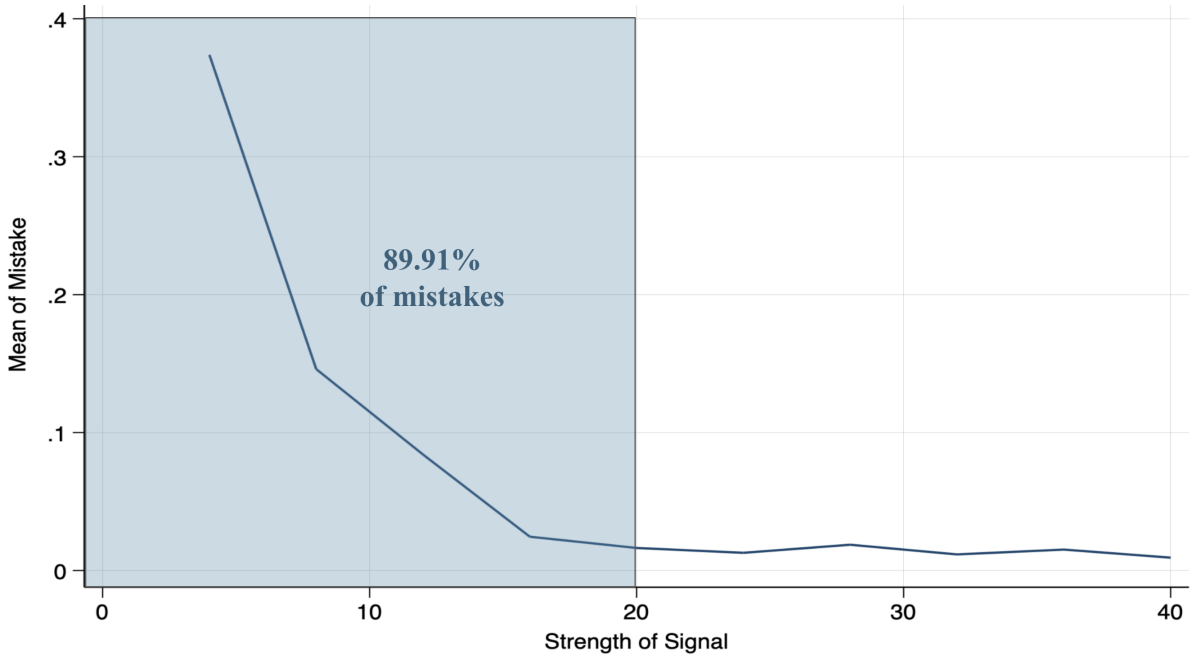

Figure 5: Mistakes By Difficulty

Due to this large inequality, I am proceeding with this analysis by including tables, graphs, and regressions at both the 40-trial level and the hardest 20-trial level. Self-deception requires ambiguity (Balcetis and Dunning, 2006; Sloman et al., 2010; Mei et al., 2023); therefore, an analysis of the hardest 20-trial level can tell us more about motivated cognition than the overall 40-trial level. Additionally, given the very low error rate of the easiest 20 trials, a comparatively larger share of mistakes could have been a result of circumstances such as accidental button presses or distraction.

Table 2: Mistakes by signal group difficulty, number and percentage.

	Correct	**Mistake**	**Total**
Hardest 20	3,319 (87.34%)	**481** (12.66%)	3,800 (100.00%)
Easiest 20	3,746 (98.58%)	**54** (1.42%)	3,800 (100.00%)
Total	7,065 (92.96%)	535 (7.04%)	7,600 (100.00%)

Table 3 summarizes the mistake rates of each group. The mistake rates of participants when faced with a win-win scenario do not vary substantially. However, when faced with the moral conflict scenario, participants displayed a wide range of error rates. The moral conflict scenario error rate of virtue consumers was 5.56% while that of the partially selfish was more that double that rate, at 11.67%. Net bias refers to the absolute difference between the error rates in moral conflict and win-win scenarios for each participant. As can be seen on the fourth column of

Table 3, net bias peaks among the partially selfish and is at its minimum among the altruists, with rates of 3.75% and -1.28% respectively. When we focus on the hardest 20 trials, the picture becomes clearer. The net bias of all groups becomes larger towards directions in line with the existence of self-deception. In the face of increased ambiguity, the selfish group exhibited a small increase of 0.22 percentage points, implying that its bias might be circumstantial. While other groups such as the partially selfish, displayed an increase of 4.58 percentage points, a 222% increase compared to the 40% level net bias. Finally, note that the overall net bias across groups is almost zero, with a value of -0.08% at the 40-trial level and -0.05% at the hardest 20-trial level. Studying the overall net bias of the participants would lead to the conclusion that there are no biases in mistake rates. Among the - self-deception facilitating - 20 hardest trial, the net bias is actually slightly lower, which would imply that there is no self-deception present. However, the individual mistake rates of the groups tell a different story. This result highlights the importance of including separation measures based on beliefs and preferences when we research biases and self-deception.

Table 3: Summary of Mistakes per Group (%)

	Mistakes	**Moral Conflict**	**Win-Win**	**(40 Trials) Net Bias**	**(20 Trials) Net Bias**
Selfish	7.61	8.18	7.05	1.14	1.36
Partially Selfish	9.79	11.67	7.92	3.75	8.33
Neutrals	8.33	9.00	7.67	1.33	2.67
Partially Altruist	6.38	6.25	6.50	-0.25	0.50
Altruist	6.42	5.78	7.06	-1.28	-2.22
Virtue Consumers	6.11	5.56	6.67	-1.11	-2.22
Total	7.04	7.00	7.08	-0.08	-0.05

Figure 6 graphically represents the mistake rates of the groups by type of scenario. As we saw in Table 3, win-win mistake rates do not exhibit a pattern, while moral conflict mistake rates are higher in the first three groups and lower in the last three. Of course, overall error rates are higher for all groups when we zoom in on the hardest 20 trials. Nevertheless, this increase in the error rate does not necessitate a directional increase in the net bias.

The net biases are amplified when we study the higher ambiguity trials, which facilitate self-deception. Figure 7 exhibits this pattern. Additionally, Figure 8 isolates the changes in the net bias of each group between the 40-trial level and the hardest 20-trial level. It must be highlighted that these are expected patterns according to the theoretical framework of this research. The changes are not uniform and are larger and directional only for the partially selfish, and neutrals, who exhibit patterns of profit-driven self-deception; and for the altruist and virtue consumers, who exhibit patterns of image-driven self-deception.

The fact that the above-mentioned pattern is observed not only implies that there might be

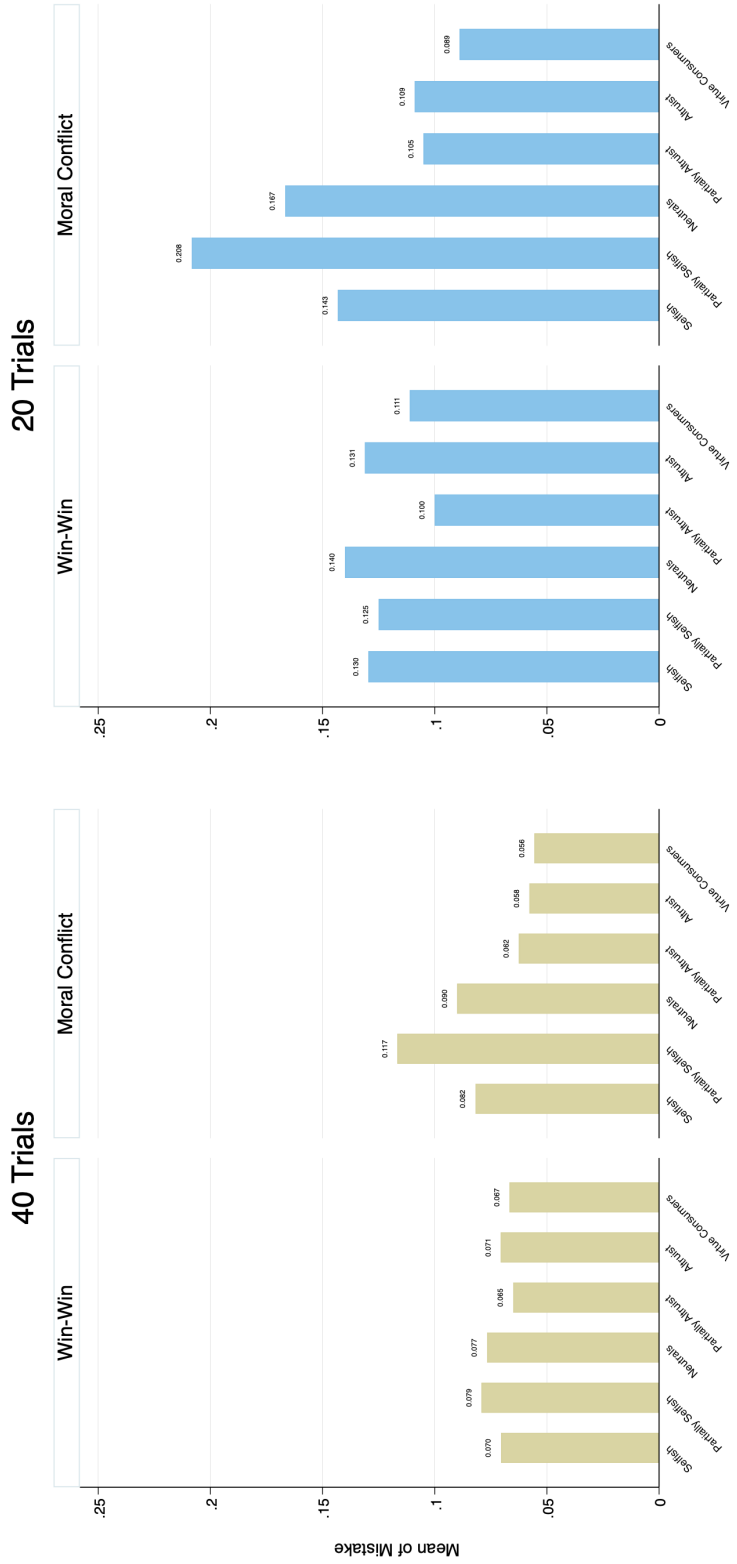

Figure 6: Type of Mistake by Group, and 40 and 20 trials

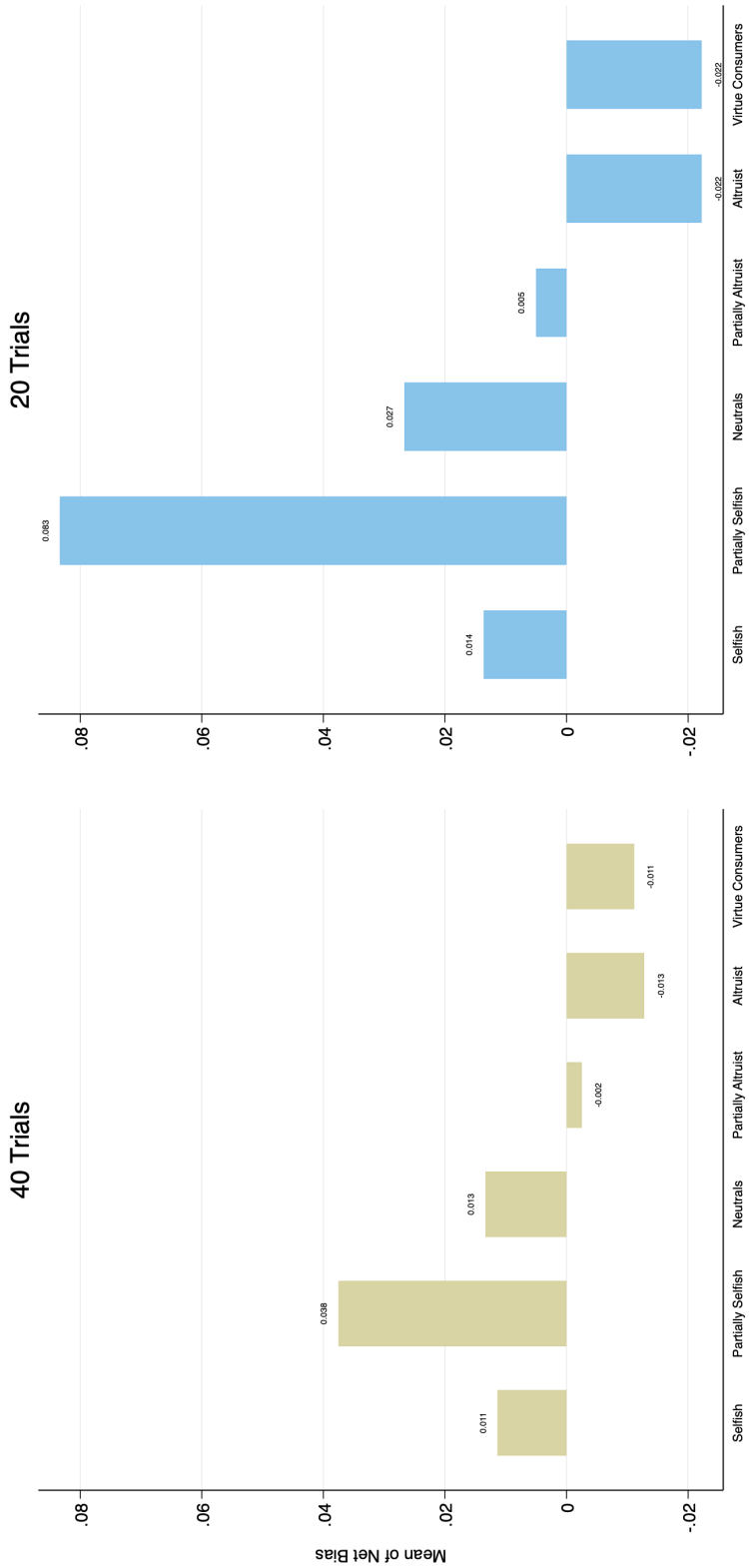

Figure 7: Net Bias by Group, 40 and 20 trials

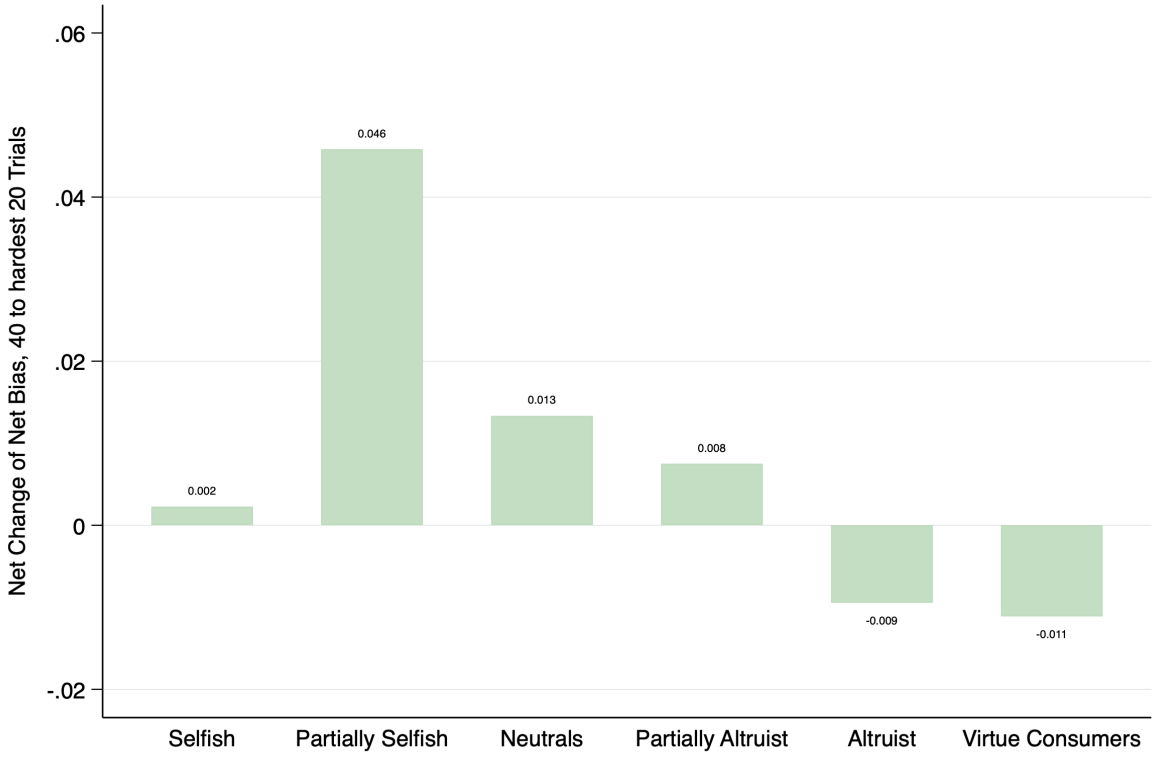

Figure 8: Net Change of Bias by Group, 40 to 20 trials

self-deception, but also that the use of Φ as a proxy of altruistic preferences and self-image is effective. Figure A2 includes a breakdown of bias direction per altruistic preference group in the 20 hardest trials (see tables in Appendix A3 for figure including all 40 trials). A negative net bias is characterized as image-driven self-deception and a positive one as profit-driven. The middle bars refer to participants who did not exhibit a bias in their mistake direction. A comparison of the image and profit columns of the figure reveals that the bias direction of participants matched the model's predictions. The imbalance between image and profit bias, in favor of the latter, begins already among the selfish and continues with the partially selfish and neutrals. Starting with the partially altruist, the imbalance is reversed and a bias towards the morally conflicting scenario is favored. The bias implying image-driven self-deception continues to be favored among the altruists and its dominance peaks among the virtue consumers. The largest difference is found in the partially selfish. Half of the partially selfish exhibited a bias towards the win-win scenario, while only 8.33% of them displayed a bias towards the morally conflicting scenario.

Reaction time can be a complicated topic to study when it comes to self-deception. Although there is evidence that less information is needed to reach a favorable conclusion Ditto and Lopez (1992), in the case of this experiment, the amount of information received was predetermined and presented to the participants for a predetermined amount of time. Figure 10 shows the net speed of making a correct or wrong scenario identification. Making a mistake generally

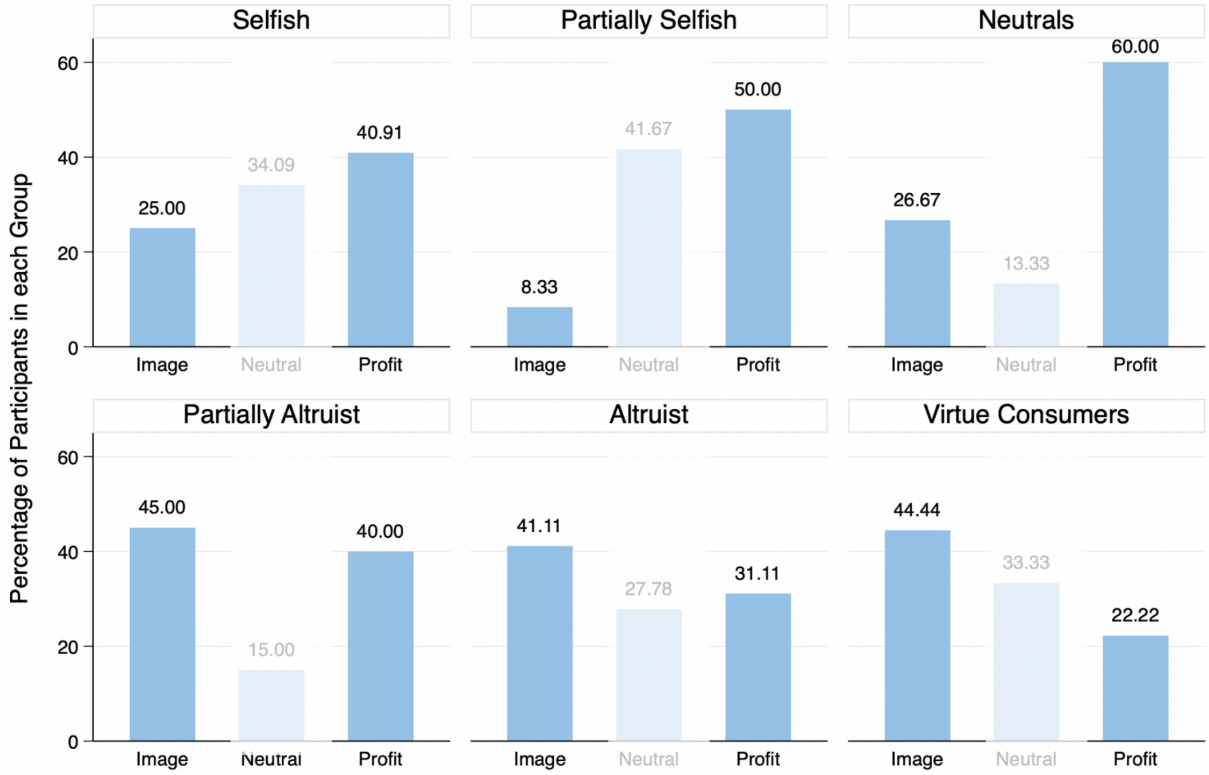

Figure 9: Share of Participants in each Category of Self-Deception - 20 Hardest Trials

takes longer, since the trials with the most mistakes are also the harder trials. In the figure, when the value of a bar is positive it signifies that the group it refers to took more time to identify, correctly or incorrectly, the win-win scenario. When the value is negative, the group of participants spent more time to identify the moral conflict scenario compared to the win-win scenario. The reaction time pattern matches the net bias pattern both for correct and incorrect identifications. More specifically, the partially selfish group took almost 7 seconds longer to mistakenly identify win-win as the correct scenario, while the virtue consumers spent almost 5 extra seconds on average when identifying the moral conflict scenario. These patterns imply a possible self-convincing in the decision-making process. Figure 11 includes only the net speed of the incorrect identifications of each group, showing that the reaction time differences of the partially selfish and virtue consumers are non-zero at their respective 95% confidence intervals.

The above results do not provide definite proof of self-deception. Nevertheless, the coexistence of these independent patterns provides significant evidence of the existence of a non-monotonic self-deception.

5.2 Regressions and Margins

Before I continue with the main econometric model of this paper, it is useful to study the aggregate patterns. Table 4 includes four different aggregate OLS models with robust standard errors and with net bias as the dependent variable. The interpretation of the table is a bit complicated, since the coefficients of each Φ group express the difference with the mean net

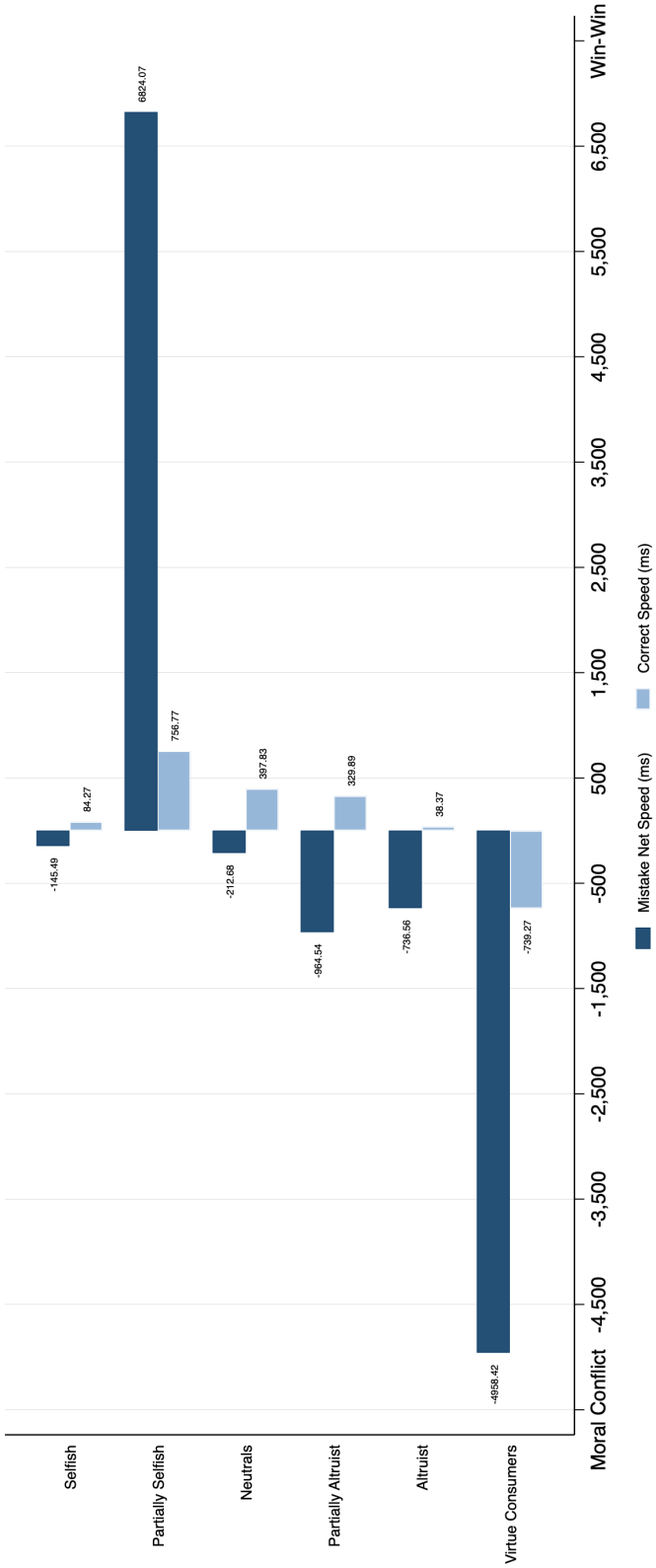

Figure 10: Net Speed 20 Trials

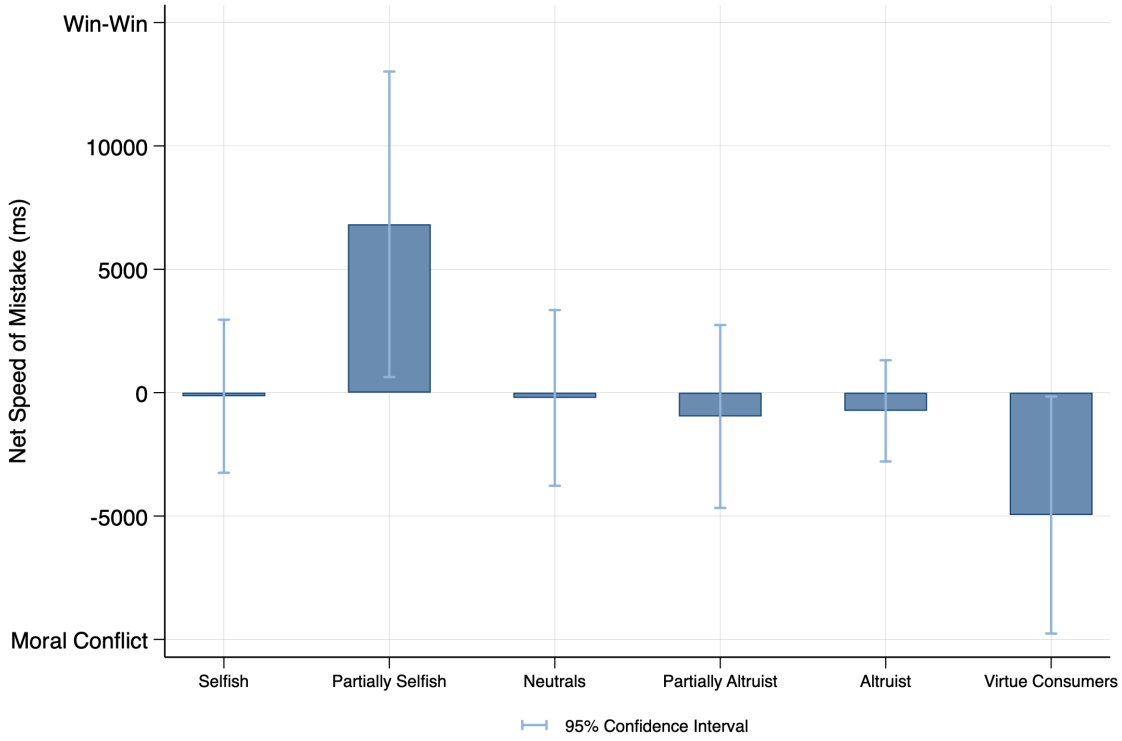

Figure 11: Net Speed Mistake (ms) with 95% CI

bias of the selfish group. The altruists in the 40-trial model without gender (model 1) show on average a lower net bias than the selfish group at $p < 0.1$. However, this limited result disappears in the rest of the models. In the 20-trial models, the partially selfish exhibit a higher net bias than the selfish, also at $p < 0.1$.

To get a better understanding of the bias size and direction of each group, we need to examine the predictive margins of the model. The 40-trial model of Table 5 shows that the partially selfish group has on average a net bias of 3.9% at a statistical significance level of just below 95%. However, the 20-trial model shows that in the harder portion of the experiment, partially selfish participants displayed an average net bias of 8.6% at $p=0.013$. Once again, the partially selfish group is the group most consistent with profit-driven self-deceptive behavior. The graphical representation of the the predictive margins can be seen on Figure 12.

Four random effects probit regressions were conducted to study self-deception at the individual trial level (Table 7). Models 1 and 2 include all 40 trials and models 3 and 4 include only the 20 hardest trials. Models 2 and 4 also include gender and timing as control variables. Most gender coefficients are statistically significant and negative due to the higher error rates of the females in the win-win scenario, which is the baseline of the gender category. The higher bias of female participants towards the morally conflicting scenario reduces the coefficients and increases the standard errors of the altruists and virtue consumers, implying that their biases towards the moral conflict scenario might be explained by gender and not their self-image con-

Table 4: Aggregate Regression Results (OLS)

	(1) 40 Trials	(2) 40 Trials	(3) 20 Trials	(4) 20 Trials
Partially Selfish	0.026	0.030	**0.070***	**0.076***
	(0.02)	(0.02)	(0.04)	(0.04)
Neutrals	0.002	0.005	0.013	0.018
	(0.02)	(0.02)	(0.04)	(0.04)
Partially Altruist	-0.014	-0.013	-0.009	-0.007
	(0.02)	(0.02)	(0.04)	(0.05)
Altruist	**-0.024***	-0.021	-0.036	-0.030
	(0.01)	(0.01)	(0.03)	(0.03)
Virtue Consumers	-0.022	-0.015	-0.036	-0.023
	(0.02)	(0.02)	(0.03)	(0.03)
Gender		✓		✓
Constant	0.011	0.003	0.014	-0.002
	(0.01)	(0.01)	(0.02)	(0.03)
R^2	0.04	0.05	0.04	0.05
N	190	190	190	190

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Predictive Margins - Aggregate OLS Model - 40 Trials

	Margin	**S.E.**	**t**	**P-value**	**[95% C.I.]**
Selfish	0.009	0.011	0.790	0.430	-0.014, 0.032
Partially Selfish	**0.039**	**0.020**	**1.900**	**0.059**	**-0.002, 0.079**
Neutrals	0.014	0.014	0.980	0.328	-0.014, 0.042
Partially Altruist	-0.004	0.022	-0.180	0.858	-0.046, 0.039
Altruist	-0.012	0.008	-1.500	0.134	-0.028, 0.004
Virtue Consumers	-0.006	0.017	-0.370	0.711	-0.040, 0.027

cerns. When compared to the baseline error rate of the selfish group in the win-win scenario, the partially selfish consistently show increased mistake rates when faced with the morally conflicting scenario. Their strongest increase is observed in the 20-trial model with a coefficient of 0.694 and a p< 0.01. The neutral group also has increased error rates when faced with a moral conflict scenario in models 2 and 3 (p< 0.1) and model 4 (p< 0.05). Finally, the selfish groups show higher error rates when faced with a moral conflict scenario compared to when they are faced with a win-win scenario at p< 0.1)

To better understand the results of the random effects probit model, we need to once again consult the predictive margins. The contrasts of the predictive margins for models 2 and 4 are displayed on Table 8. The contrasts refer to the difference in the probability of mistake that each group exhibits between facing a morally conflicting scenario and facing a win-win scenario. Once again, the partially selfish group is the only group with a bias that is significantly different than 0. At the 40-trial model, the bias towards the win-win scenario is of minor statistical significance (p< 0.1), while in the 20-trial model the statistical significance increases and passes the 95% threshold. The marginal bias in these two cases almost mirrors the levels observed at the aggregate model's margins in Tables 5 and 6.

Table 6: Predictive Margins - Aggregate OLS Model - 20 Trials

	Margin	**S.E.**	**t**	**P-value**	**[95% C.I.]**
Selfish	0.009	0.021	0.450	0.652	-0.032, 0.051
Partially Selfish	**0.086**	**0.034**	**2.510**	**0.013**	**0.018**, **0.153**
Neutrals	0.028	0.032	0.870	0.386	-0.035, 0.091
Partially Altruist	0.002	0.041	0.060	0.952	-0.079, 0.084
Altruist	-0.021	0.016	-1.330	0.187	-0.052, 0.010
Virtue Consumers	-0.014	0.026	-0.510	0.608	-0.065, 0.038

Finally, the linear regression results of Table 9 show that the partially selfish group took almost 7 seconds longer than the selfish group to make a mistake when faced with a moral conflict scenario ($p < 0.05$) and, compared to the same baseline, the virtue consumers took almost 5 seconds more to make a mistake when faced with a win-win scenario. The margins on Table 10 confirm the pattern observed in the regression results. On average, the partially selfish took 6.82 seconds longer to make a mistake towards the win-win scenario and the virtue consumers took 4.95 seconds longer to make a mistake towards the morally conflicting scenario. The predictive margins are visualized in Figure 14.

6 Discussion

The results of the experimental findings of this research are in line with its theoretical framework; nevertheless, the statistical strength of the evidence is weak. There is non-monotonic self-deception, with biases moving in different directions depending on the category of altruistic preferences and self-image concerns that individuals belong to. Despite the need for further experimental research, the behavior of participants in this experiment in terms of bias sign, bias magnitude, and reaction time, points to the existence of a mechanism that triggers two distinct directions of self-deception for participants with different self-image concerns.

Beyond the aggregate Much of the literature on self-deception attempts to shed light at individual-level biases. Nevertheless, to achieve this goal effectively, we need to find ways to include a measure of self-image in our models and our experimental designs. This experiment was an attempt to do that. The bias of the more selfish groups toward the win-win scenario is of roughly equal magnitude with the bias toward the morally conflicting scenario that the more altruistic groups exhibited. The average net bias across all participants was approximately zero. As a result, the omission of Φ would lead us to believe that there is no pattern of behavior.

The non-monotonic relationship between self-deception and the self-image of altruism will not always result in a near-zero average bias. In different settings, the varying self-image group dynamics could drive the net bias in either direction. Studying these patterns at the aggregate level and without the inclusion of self-image measurements will not be possible.

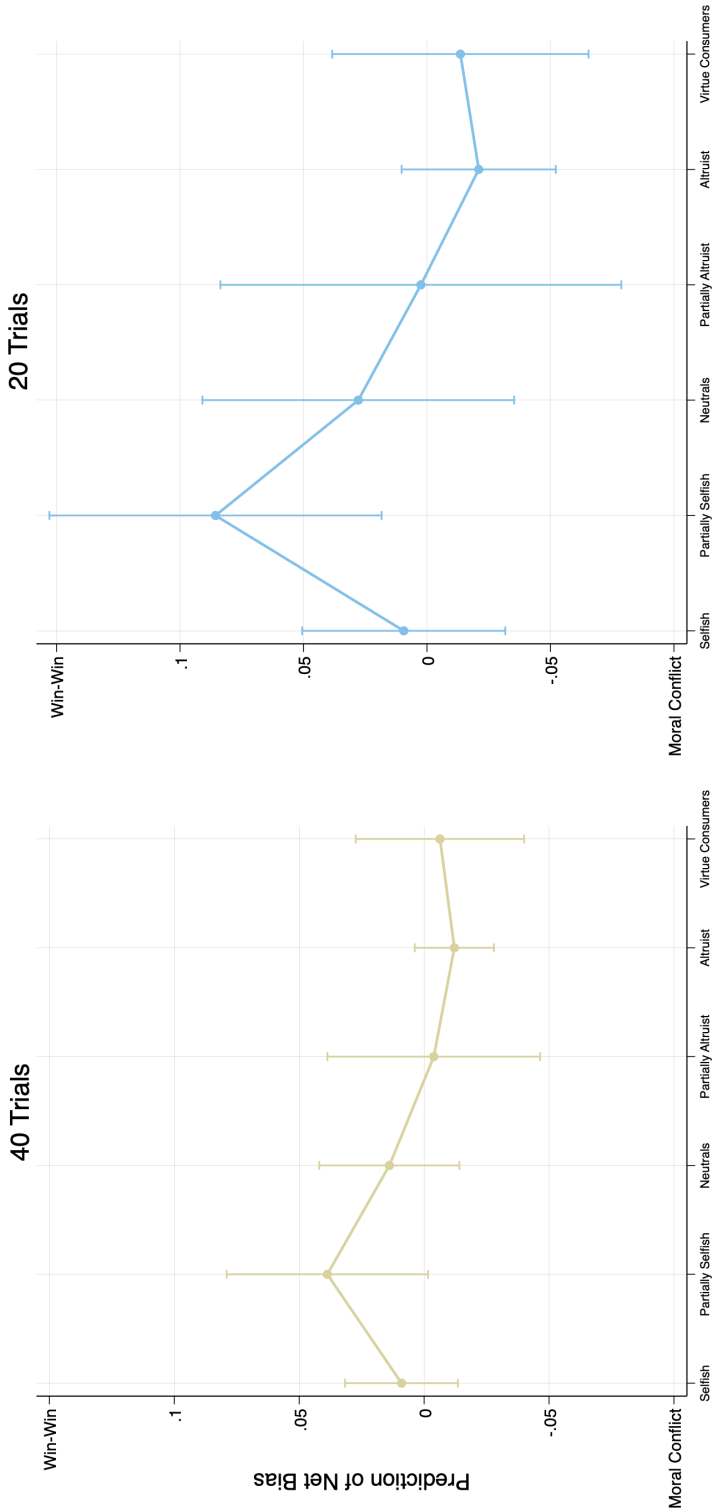

Figure 12: Predictive Margins OLS, 40 and 20 Trials

Profit and Image-Driven Self-Deception When the setting becomes more complex and motivations collide, we can have more than one type of self-deception. In this case, two types of

Table 7: Random Effects Probit Model, Probability of Mistake

	(1) 40 Trials	(2) 40 Trials	(3) 20 Trials	(4) 20 Trials
Signal Strength	-0.076***	-0.076***	-0.138***	-0.138***
	(0.01)	(0.01)	(0.01)	(0.01)
Win-Win				
#Partially Selfish	0.108	0.078	0.016	-0.013
	(0.23)	(0.24)	(0.23)	(0.24)
#Neutrals	0.079	0.031	0.062	0.001
	(0.16)	(0.16)	(0.17)	(0.17)
#Partially Altruist	0.025	0.022	-0.152	-0.153
	(0.15)	(0.15)	(0.18)	(0.18)
#Altruist	0.019	-0.019	0.034	-0.008
	(0.11)	(0.11)	(0.12)	(0.12)
#Virtue Consumers	-0.027	-0.090	-0.117	-0.182
	(0.21)	(0.23)	(0.20)	(0.21)
Moral Conflict				
#Selfish	0.129	0.302	0.103	**0.376***
	(0.10)	(0.18)	(0.12)	(0.21)
#Partially Selfish	**0.369***	**0.542****	**0.419****	**0.694*****
	(0.19)	(0.26)	(0.20)	(0.27)
#Neutrals	0.215	**0.373***	**0.268***	**0.517****
	(0.14)	(0.19)	(0.16)	(0.21)
#Partially Altruist	-0.016	0.161	-0.088	0.191
	(0.21)	(0.26)	(0.21)	(0.26)
#Altruist	-0.121	0.050	-0.123	0.144
	(0.11)	(0.18)	(0.12)	(0.20)
#Virtue Consumers	-0.154	0.010	-0.292	-0.023
	(0.32)	(0.38)	(0.27)	(0.33)
Timing		**0.006*****		0.002
		(0.00)		(0.00)
Gender		✓		✓
Constant	**-0.395*****	**-0.450*****	0.148	0.204
	(0.11)	(0.14)	(0.12)	(0.13)
$\ln(\sigma_u^2)$	-1.954***	-1.987***	-2.525***	-2.608***
	(0.19)	(0.20)	(0.32)	(0.36)
N	7600	7600	3800	3800
Groups	190	190	190	190

Standard errors in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

self-deception are identified: profit-driven and image-driven self-deception. The conflict between the two motivations - profit and self-image - exists in both types, but one of the two dominates in each case. Note that the structures of the utility functions remain the same, the magnitude of self-image concerns and altruistic preferences are what cause the change of bias direction.

Moreover, the fact that profit-driven self-deception peaks among the partially selfish and shows a positive trend even among the selfish implies that self-image concerns might begin to

Moral Conflict vs Win-Win	P-value	Contrast	Std. Err.
40 trials			
Selfish	0.3189	0.011	0.011
Partially Selfish	**0.0966**	**0.034**	**0.020**
Neutrals	0.2155	0.017	0.013
Partially Altruist	0.7856	-0.006	0.022
Altruist	0.1138	-0.012	0.008
Virtue Consumers	0.5794	-0.009	0.016
Joint	0.2307		
20 trials			
Selfish	0.5282	0.013	0.021
Partially Selfish	**0.0198**	**0.079**	**0.034**
Neutrals	0.2186	0.040	0.032
Partially Altruist	0.8696	0.007	0.041
Altruist	0.1391	-0.023	0.015
Virtue Consumers	0.4060	-0.019	0.022
Joint	0.1106		

Table 8: Contrasts of Predictive Margins, 40 and 20 Trials

play a role at very low levels of altruism. A less statistically consistent result is the image-driven self-deception of the more altruistic groups and implies that some individuals are overestimating the extent of a problem - along with the appropriate personal cost - to increase their self-image. More importantly, in this experiment, the self-deception of those individuals not only decreases their payout but also the payout of the charity. Each time the image self-deceivers mistakenly saw a conflict scenario, they chose the option that gives more to the charity under the conflict scenario; however, in the correct, win-win, scenario the same choice results in a lower payout to the charity.

The two directions of self-deception Once we establish the existence of the two self-deception directions, we can see it in a variety of contexts. In this experiment, people who know that they might betray their self-image avoided the conflicting scenario by deploying profit-driven self-deception. However, when altruistic preferences start taking over, bias lowers. After a point, the self-image value of feeling like an altruist makes the diagnostic utility of acting altruistically in the conflict scenario too attractive, resulting in image-driven self-deception among the more altruistic groups. My experimental results also fit the theoretical work performed by Nyborg (2011), which separates consumers into duty-oriented and warm glow consequentialist. The

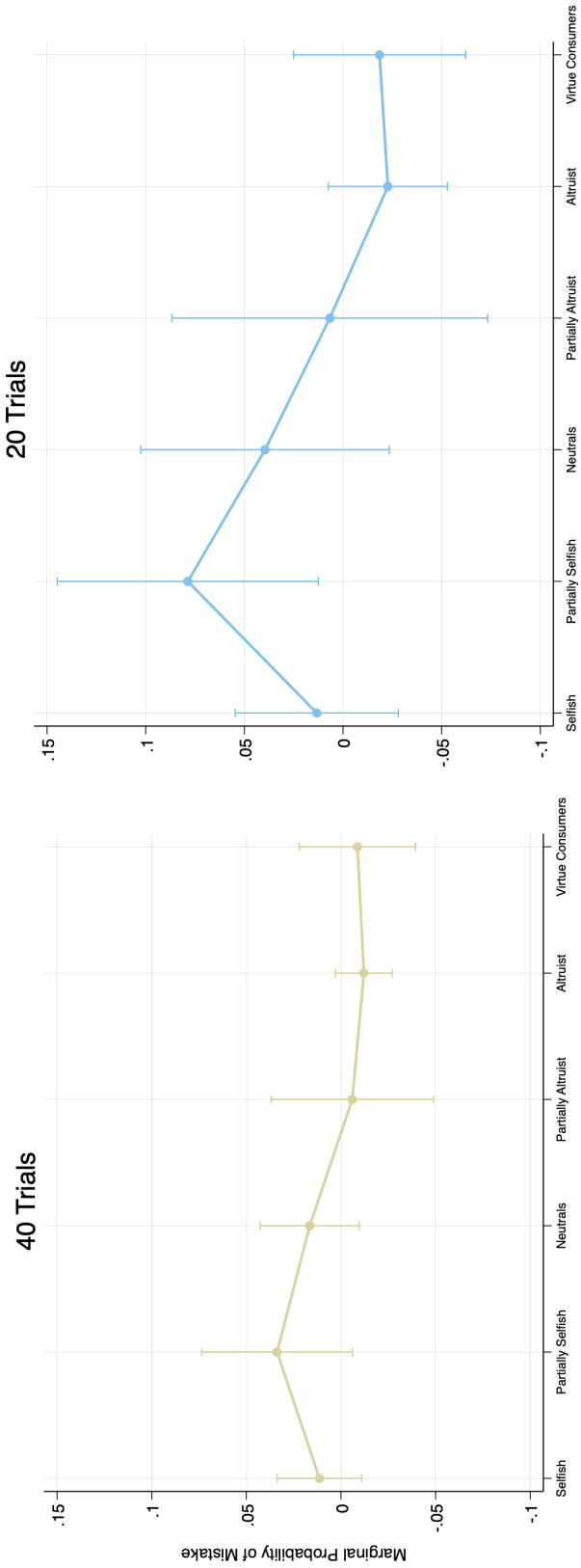

Figure 13: Categorical Contrasts 40-20

Table 9: Net Reaction Time of Mistake, OLS

	Mistake Net Reaction Time
Partially Selfish	**6969.559****
	(3105.182)
Neutrals	-67.191
	(2233.031)
Partially Altruist	-819.054
	(2329.755)
Altruist	-591.074
	(1860.614)
Virtue Consumers	**-4812.933****
	(2472.348)
Constant	-145.488
	(1540.964)
R-squared	0.0536
N	174

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

	Margin	**S.E.**	**t**	**P-value**	**[95% C.I.]**
Selfish	-145	1540	-0.090	0.925	-3187, 2896
Partially Selfish	**6824**	**2695**	**2.530**	**0.012**	**1501, 12146**
Neutrals	-212	1616	-0.130	0.895	-3403, 2977
Partially Altruist	-964	1747	-0.550	0.582	-4414, 2485
Altruist	-736	1042	-0.710	0.481	-2795, 1322
Virtue Consumers	**-4958**	**1933**	**-2.560**	**0.011**	**-8775, -1141**

Table 10: Predictive Margins - Net Mistake Speed - OLS - 20 Trials

former group avoids information, even at a cost, if they believe it will increase their moral obligations. The latter group seeks information, driven by a desire to enhance their self-image.

Despite being driven by the same type of motivations individuals might behave very differently. For example, profit-driven self-deceivers might ignore the dangers of climate change, while image-driven self-deceivers might be putting their hopes on individually costly solutions that do not address the problem sufficiently. As employees, profit-driven self-deceivers might ignore their company's environmentally harmful actions and the second group might fall victim to company greenwashing. Both of these problems are caused by an effort to manipulate one's self-image. Ambiguity and a lack of transparency, in the first case, do not allow us to benefit from the potential - albeit limited - contribution of the profit-driven self-deceivers. In the second case, a group willing to contribute significantly to the cause of climate change is inefficiently utilized. On the bright side, identifying the kind of self-deception individuals deploy can also create opportunities. Profit-driven self-deceivers can be seen as a group that can be convinced with

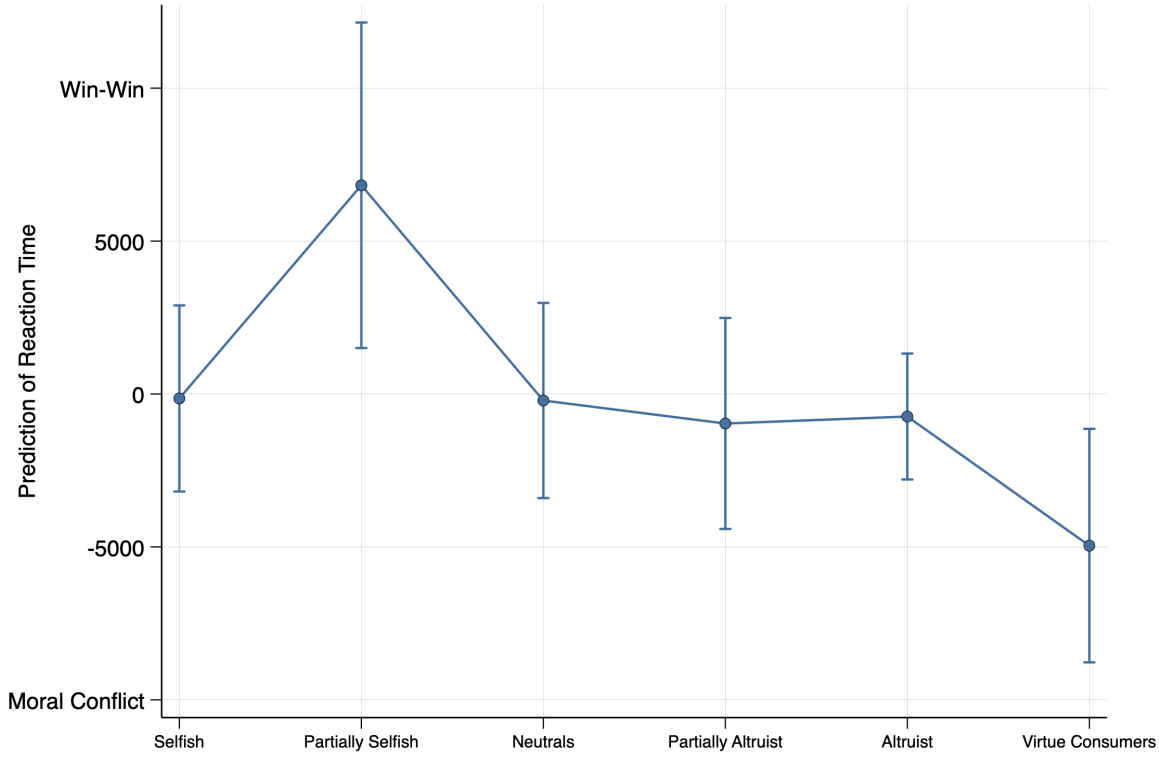

Figure 14: Net Speed of Mistake, Margins

more robust information, while the willingness and resources of the image-driven self-deceivers can be stirred towards more productive solutions.

Political correctness is a topic where the profit and image-driven analysis of self-deception can help us understand some behaviors that limit our ability to communicate and progress. Political correctness is not by itself a negative term. Nobody desires to live in a world of constant offense. The profit-driven self-deception equivalent here is the person who ignores clear opportunities to speak up on important offenses. This person might allow individuals around her (or herself) to be hurt due to their self-deception. This person, of course, does not need to only exist in the sphere of political correctness. For example, a man who stands only to lose socially and professionally, but cares about equality to an extent, might self-deceive away clear injustices against female coworkers in the workspace. On the other hand, many people hold a disdain for political correctness because of extreme behaviors and phenomena such as cancel culture. Overly sensitive political correctness is the definition of image-driven self-deception. The person who gets offended on behalf of some other person in response to a comment that most would deem innocent is an image-driven self-deceiver.

Limitations and future research The inclusion of $\Phi(\alpha, \sigma)$ as a proxy for the self-image of altruism proved far from perfect, but was a good first step toward the measurement of self-image. The overall patterns and the fact that the directions of net bias coincided with the model predictions are encouraging. Nevertheless, future research should come up with approaches to

solve the problem of self-image measurement. Clarity in the measurement of self-image should be a priority.

The experimental design of this research had a few drawbacks. Many of the signal levels were too easy, which significantly decreased the statistical power of the results. In the future, a more efficient approach would be to skip signal levels and repeat the high error levels instead. Ambiguity is very important when it comes to self-deception and this experimental design did not include enough of it. If, similarly to the present research, an implicit measurement of self-image is needed, the stronger signal trials can be included but the trials in between need not be.

7 Conclusion

This research provides novel insights into the complex relationship between self-deception, altruism, and self-image concerns. The experimental findings highlight the existence of two distinct types of self-deception: profit-driven and image-driven. Profit-driven self-deception is characterized by individuals underestimating altruistic opportunities to protect their material self-interest, while image-driven self-deception involves overestimating scenarios that require ethical judgment to enhance one's self-image.

This paper was an attempt to quantify self-image, but the measurement used is only a proxy; therefore, future research on the topic must improve on the measurement of self-image. Exploring more complex and motivational conflicting settings and testing its findings on real-world applications should be at the forefront of self-deception research.

The non-monotonic relationship between self-deception and self-image of altruism suggests that individuals with different levels of altruistic preferences and self-image concerns exhibit opposing biases in their decision-making processes. This divergence underscores the importance of considering individual differences in self-image when studying self-deception.

One significant implication of these findings is that self-deception is not an effortless process. The reaction time patterns observed in the experiment indicate that participants engaged in a self-convincing process, which required time and cognitive effort. If self-deception requires effort, it might be easier to find tools to combat it.

Future research should focus on refining the measurement of self-image and exploring more complex, motivationally conflicting settings. Understanding the nuances of self-deception can have practical applications in various domains, including health behaviors, conflict of interest scenarios, and prosocial behavior. By identifying the types of self-deception individuals are prone to, interventions can be tailored to address specific biases, leading to more effective strategies for promoting honesty, ethical behavior, and better decision-making.

In conclusion, this study advances our understanding of self-deception by highlighting the role of self-image concerns and the non-monotonic nature of the relationship between self-deception and altruism. The experimental design and findings contribute to the broader literature on belief-based utility and motivated cognition, offering a novel framework for future research.

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A Appendix

A.I Rejection due to Utility Violations

Between Session 2 and Session 3, the experiment moved from the experimental laboratory at Universitat Pompeu Fabra (BESLAB) to the one at the University of Barcelona. Due to an omission in the recruitment process, a few participants with poor English language skills were included in Session 3. As a result, the utility violations were significantly higher than the rest of the groups (Table A1). A utility violation is defined as an action that reduced both the participant's own payoff and the one of the charity with the knowledge of the participant.

Table A1: Comparison of Violations by Session

Session No.	**1**	**2**	**3**	**4**
2	-0.004104	.	.	.
	(1.000)	.	.	.
3	0.031213	0.035317	.	.
	(0.002)	**(0.003)**	.	.
4	0.0055	0.009604	-0.025713	.
	(1.000)	(1.000)	**(0.073)**	.
5	-0.006226	-0.002122	-0.03744	-0.011726
	(1.000)	(1.000)	**(0.005)**	(1.000)

P-values in parentheses

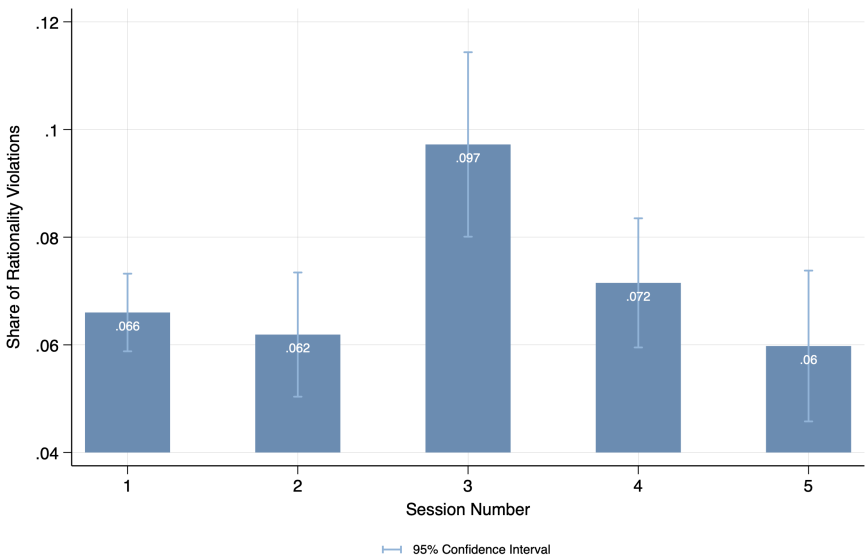

Figure A1: Violations By Session

A.II Tables

Table A2: Mistakes by Gender, number and percentage

	Correct	**Mistake**	**Total**
Female	4,760 (93.55%)	328 (6.45%)	5,088 (100.00%)
Male	3,621 (94.30%)	219 (5.70%)	3,840 (100.00%)
Other	175 (91.15%)	17 (8.85%)	192 (100.00%)
Total	8,556 (93.82%)	564 (6.18%)	9,120 (100.00%)

Table A3: Actions by Gender, number and percentage

	A	**B**	**Total**
Female	3,258 (64.03%)	1,830 (35.97%)	5,088 (100.00%)
Male	2,738 (71.30%)	1,102 (28.70%)	3,840 (100.00%)
Other	136 (70.83%)	56 (29.17%)	192 (100.00%)
Total	6,132 (67.24%)	2,988 (32.76%)	9,120 (100.00%)

Table A4: Field of Study by Gender

	Female	**Male**	**Other**	**Total**
Communication	1			1
Economics & Business	63	58		121
Engineering	3	8	1	12
History	1			1
Natural Sciences	15	6		21
Other	16	4	1	21
Social Sciences	7	4	2	13
Total	106	80	4	190

A.III Figures

Figure A2: Share of Participants in each Category of Self-Deception